Annex to certificate

Parker Hannifin 080228 P0001 C01.11

List of covered Solenoid Valves types F, V, X and N

Customer:

Parker Hannifin Manufacturing SRL
Corsico (MI)
ITALY

Contract No.: Parker Hannifin 0802-28-C

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Purpose and scope

This annex to the certificate Parker Hannifin 080228 P0001 C01.11 contains the listing of all the certified combinations of the Solenoid Valves types F, V, X and N together with their electrical parts.

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1 Type overview with failure rates according to IEC 61508:2010

Profile 3: Mechanical field products have minimal self heating and are subjected to daily temperature swings.

"U" present in front of the valve codification stands for NPT connections (e.g. U133X5196), whereas "U" absent stands for BPS connections (e.g. 133X5196).

Valves series F & V

Solenoid			Failure Rate in FIT [:= 10^{-9}/h] Profile 3	
Valve Body	**Electrical Part**	**Power**	$\square_{safe}$	$\square_{dang}$
U131**F**5695 U131**F**5295 U121**V**5595 U121**V**7595 U133**V**5595 U133**V**7595 U133**V**5695 U133**V**7695	492965	2.3W	150	66
	496565	2.3W	150	66
	492210	2.3W	150	66
	492310	6W	107	66
	496555	6W	107	66
	496700	6W	107	66
	496560	8W	107	66
	496800	8W	107	66
U121**V**5596	492310	6W	107	66
	496555	6W	107	66
	496700	6W	107	66
	496560	8W	107	66
	496800	8W	107	66

Valves series X

Solenoid			Failure Rate in FIT [:= 10^{-9}/h] Profile 3	
Valve Body	**Electrical Part**	**Power**	$\square_{safe}$	$\square_{dang}$
U**131X**1201 (U)**131X**1101	492965	2.3W	119	168
	496565	2.3W	119	168
	492210	2.3W	119	168
	492310	6W	76	168
	496555	6W	76	168
	496700	6W	76	168
	496560	8W	76	168
	496800	8W	76	168
U**133X**5156 U**133X**5195 U**133X**7195 U**133X**5196 U**133X**5296 U**133X**7296 (U)**133X**01 U**133X**0111 U**133X**S9443	492965	2.3W	119	168
	496565	2.3W	119	168
	492210	2.3W	119	168
	492310	6W	76	168
	496555	6W	76	168
	496800	8W	76	168
	492965	2.3W	119	168
	496565	2.3W	119	168
U**133X**7156 U**133X**7196 U**133X**7759 U**133X**7709	492965	2.3W	119	168
	496565	2.3W	119	168
	492210	2.3W	119	168
	492310	6W	76	168
	496555	6W	76	168
	496700	6W	76	168
	497105	8W	76	168
	496560	8W	76	168
	496800	8W	76	168
131X1131	492310	6W	76	168
	496555	6W	76	168
	496700	6W	76	168
	496560	8W	76	168
	496800	8W	76	168

Solenoid			Failure Rate in FIT [:= 10^{-9}/h]	
			Profile 3	
Valve Body	**Electrical Part**	**Power**	λ_{safe}	λ_{dang}
U**133X**5152 U**133X**5192	492310	6W	76	168
	496555	6W	76	168
	496700	6W	76	168
	496560	8W	76	168
	496800	8W	76	168
U**033X**0111 U**033X**5152 U**033X**5156 U**033X**5256 U**033X**5195 U**033X**7195 U**033XS**9443	482870	3W	76	168
	492310	6W	76	168
	496555	6W	76	168
	496700	6W	76	168
	496560	8W	76	168
	496800	8W	76	168
U**033X**7156 U**033X**7759	492310	6W	76	168
	496555	6W	76	168
	496700	6W	76	168
	497105	8W	76	168
	496560	8W	76	168
	496800	8W	76	168
U**331X**2309	492965	2.3W	119	200
	496565	2.3W	119	200
	492310	6W	76	200
	496555	6W	76	200
	496700	6W	76	200
	497105	8W	76	200
	496560	8W	76	200
	496800	8W	76	200

Valves series N

Solenoid			Failure Rate in FIT [:= 10^{-9}/h]	
			Profile 3	
Valve Body	Electrical Part	Power	λ_{safe}	λ_{dang}
341 **N**31 341 **N**32	483371	8W	107	654
	495905	8W	107	654
	481000	8W	107	654
	481865	8W	107	654
	482725	8W	107	654
	496110	9W	107	654
	483510	9W	107	654
341 **N**32	496155	14W	332	654
341 **N**3102	483371	8W	107	654
	481000	8W	107	654
	481865	8W	107	654
	482725	8W	107	654
341 **N**3108	495905	8W	107	654
	481000	8W	107	654
	481865	8W	107	654
	482725	8W	107	654
341 **N**3130	495905	8W	107	654
	481865	8W	107	654
	482725	8W	107	654
341 **N**3190 341 **N**3290	483580..	2.3W	107	654
	483960..	2.3W	107	654
	488650..	3W	107	654
	488660..	3W	107	654
341 **N**3197 341 **N**3297	496125	1.6W	57	654
	482740	1.6W	57	654
	482745	1.6W	57	654
341 **N**3197	495910	2.3W	150	654
341 **N**3197	495900	2.5W	107	654

2 Type overview with corresponding failure rates and partial valve stroke testing (PVST)

Partial Valve Stroke testing of the SIF provides a full cycle test of the solenoid valve. Further information about PVST can be found in the draft technical report ISA-TR96.05.01-200_. As the diagnostic coverage (DC) depends on the quality of the PVST (time measurement) for certain failure modes only a DC of 75% was assumed.

Profile 3: Mechanical field products have minimal self heating and are subjected to daily temperature swings.

"U" present in front of the valve codification stands for NPT connections (e.g. U133X5196), whereas "U" absent stands for BPS connections (e.g. 133X5196).

Valves series F & V

Solenoid			Failure Rate in FIT [:= 10^{-9}/h] Profile 3		
Valve Body	**Electrical Part**	**Power**	λ_{safe}	λ_{dd}	λ_{du}
U131**F**5695 U131**F**5295 U121**V**5595 U121**V**7595 U133**V**5595 U133**V**7595 U133**V**5695 U133**V**7695	492965	2.3W	150	61	5
	496565	2.3W	150	61	5
	492210	2.3W	150	61	5
	492310	6W	107	61	5
	496555	6W	107	61	5
	496700	6W	107	61	5
	496560	8W	107	61	5
	496800	8W	107	61	5
U121**V**5596	492310	6W	107	61	5
	496555	6W	107	61	5
	496700	6W	107	61	5
	496560	8W	107	61	5
	496800	8W	107	61	5

Valves series X

Solenoid			Failure Rate in FIT [:= 10^{-9}/h]		
			Profile 3		
Valve Body	**Electrical Part**	**Power**	λ_{safe}	λ_{dd}	λ_{du}
U**131X**1201 (U)**131X**1101	492965	2.3W	119	137	31
	496565	2.3W	119	137	31
	492210	2.3W	119	137	31
	492310	6W	76	137	31
	496555	6W	76	137	31
	496700	6W	76	137	31
	496560	8W	76	137	31
	496800	8W	76	137	31
U**133X**5156 U**133X**5195 U**133X**7195 U**133X**5196 U**133X**5296 U**133X**7296 (U)**133X**01 U**133X**0111 U**133X**S9443	492965	2.3W	119	137	31
	496565	2.3W	119	137	31
	492210	2.3W	119	137	31
	492310	6W	76	137	31
	496555	6W	76	137	31
	496700	6W	76	137	31
	496560	8W	76	137	31
	496800	8W	76	137	31
	497105	8W	76	137	31
U**133X**7156 U**133X**7196 U**133X**7759 U**133X**7709	492965	2.3W	119	137	31
	496565	2.3W	119	137	31
	492210	2.3W	119	137	31
	492310	6W	76	137	31
	496555	6W	76	137	31
	496700	6W	76	137	31
	497105	8W	76	137	31
	496560	8W	76	137	31
	496800	8W	76	137	31
131X1131	492310	6W	76	137	31
	496555	6W	76	137	31
	496700	6W	76	137	31
	496560	8W	76	137	31
	496800	8W	76	137	31
U**133X**5152 U**133X**5192	492310	6W	76	137	31
	496555	6W	76	137	31
	496700	6W	76	137	31
	496560	8W	76	137	31
	496800	8W	76	137	31

Solenoid			Failure Rate in FIT [:= 10^{-9}/h]		
			Profile 3		
Valve Body	Electrical Part	Power	$\square_{safe}$	$\square_{dd}$	$\square_{du}$
U033X0111 U033X5152 U033X5156 U033X5256 U033X5195 U033X7195 U033XS9443	482870	3W	76	137	31
	492310	6W	76	137	31
	496555	6W	76	137	31
	496700	6W	76	137	31
	496560	8W	76	137	31
	496800	8W	76	137	31
U033X7156 U033X7759	492310	6W	76	137	31
	496555	6W	76	137	31
	496700	6W	76	137	31
	497105	8W	76	137	31
	496560	8W	76	137	31
	496800	8W	76	137	31
U331X2309	492965	2.3W	119	163	37
	496565	2.3W	119	163	37
	492310	6W	76	163	37
	496555	6W	76	163	37
	496700	6W	76	163	37
	497105	8W	76	163	37
	496560	8W	76	163	37
	496800	8W	76	163	37

Valves series N

Solenoid			Failure Rate in FIT [:= 10^{-9}/h]		
			Profile 3		
Valve Body	Electrical Part	Power	$\square_{safe}$	$\square_{dd}$	$\square_{du}$
341 **N31** 341 **N32**	483371	8W	107	507	147
	495905	8W	107	507	147
	481000	8W	107	507	147
	481865	8W	107	507	147
	482725	8W	107	507	147
	496110	9W	107	507	147
	483510	9W	107	507	147
341 **N32**	496155	14W	332	507	147
341 **N3102**	483371	8W	107	507	147
	481000	8W	107	507	147
	481865	8W	107	507	147
	482725	8W	107	507	147
341 **N3108**	495905	8W	107	507	147
	481000	8W	107	507	147
	481865	8W	107	507	147
	482725	8W	107	507	147
341 **N3130**	495905	8W	107	507	147
	481865	8W	107	507	147
	482725	8W	107	507	147
341 **N3190** 341 **N3290**	483580..	2.3W	107	507	147
	483960..	2.3W	107	507	147
	488650..	3W	107	507	147
	488660..	3W	107	507	147
341 **N3197** 341 **N3297**	496125	1.6W	57	507	147
	482740	1.6W	57	507	147
	482745	1.6W	57	507	147
341 **N3197**	495910	2.3W	150	507	147
341 **N3197**	495900	2.5W	107	507	147

3 Status of the document

3.1 Releases

Version History:	V1, R0:	Initial version 24 September 2009
	V1, R1:	Updated after review, 27 October 2009
	V1, R2:	Electrical part number 497105 added April 2014
	V2, R0:	Updated to IEC 61508:2010 September 2014
	V2, R1:	Minor typo corrected.
	V2, R2:	New certificate revision January 2015
	V2, R3:	Typos corrected February 2015
	V4, R0:	Company name change November 2017
	V4, R1:	Added valves U331X2309, U033X0111, U133X7759, U133X7709 and U033X7759. 1D variants removed as obsolete, 16 Apr 2019.
	V4, R2:	Coils 496555 and 496700 added to U331X2309, 18 Jun 2019.
	V5, R0:	Updated after surveillance audit 19 Feb 2020.

Author:	Peter Söderblom	
Review:	V1, R0	Audun Opem, Certifying assessor, Customer
	V1, R2	Dr. Cornelius Rieß, Certifying assessor
	V2, R0	Dr. Cornelius Rieß, Certifying assessor
	V4, R0	Customer
	V5, R0	Steven Close, Certifying assessor

Release status: Released